

Armaan Malhotra

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SUMMARY

Engineering undergraduate building a foundation in cybersecurity through self-directed, hands-on training and CTFs. Learning web application security, network forensics, and Linux-based security tooling via structured platforms (TryHackMe, PortSwigger Web Security Academy, OverTheWire), with growing practical exposure to Burp Suite, Wireshark, and Metasploit in lab environments.

EDUCATION

Thapar Institute of Engineering & Technology

Bachelor of Engineering, Electronics and Computer Engineering

Patiala, Punjab

Aug. 2025 – Present

TECHNICAL SKILLS

Languages: C, SQL

Security Tools: Kali Linux, Burp Suite, Wireshark, Nmap, Metasploit, Nessus, Searchsploit, Ghidra, CyberChef, Steghide, John the Ripper, Cisco Packet Tracer

Developer Tools: Git, GitHub, Docker, VS Code, Figma, Canva

Concepts: Network Security, TCP/IP, DNS Resolution, Subnetting, Routing & Switching, Vulnerability Assessment, Web Application Security, OWASP Top 10, SQL Injection, WAF Bypass, Out-of-Band Exfiltration, SSTI, AES-ECB, Steganography, PCAP Analysis, Digital/Disk Forensics, XOR Cryptanalysis, Cryptographic Hashing, Linux Privilege Escalation

CERTIFICATIONS & COURSES

Complete Ethical Hacking Bootcamp: Zero to Mastery (Udemy) | EC-Council Cybersecurity Workshop: Steganography & Network Forensics (1 CPE Credit) | XPLO(IT)2 CTF – Certificate of Participation, OWASP (2025)

EXPERIENCE

Self-Directed Cybersecurity Foundations Program

June 2026 – Present

160+ hours of structured, deliberate practice

- Completed TryHackMe's DNS in Detail, HTTP in Detail, Linux Fundamentals, Linux Strength Training, and Bash Scripting rooms; analyzed live network traffic in Wireshark and automated tasks with Bash and GPG
- Cracked hashes with John the Ripper and cleared 10 levels of OverTheWire's Bandit wargame, practicing Linux privilege escalation and SSH-based exploitation
- Practiced offensive web application testing across PortSwigger's 18-lab SQL Injection curriculum and TryHackMe's Burp Suite: Repeater room
- Bypassed WAF filters and extracted data via blind/UNION-based and out-of-band exfiltration techniques – the same attack classes a SOC analyst is trained to detect

PROJECTS

AURA – Autonomous Understanding & Response Agent | *Python, FastAPI, React, XGBoost, spaCy*

Apr. 2026

- Collaborated in a 4-person team as backend/API engineer to build an 8-stage real-time AI emergency response system for ET GEN AI Hackathon 2026
- Engineered FAM (spaCy-based medical entity extraction) and ECHO (19-dimensional panic scoring via XGBoost) to adapt responses to the user's cognitive state
- Built a Medical Rule Engine validating AI-generated first-aid instructions against 30+ safety rules, with a full audit trail API logging every decision

CruxHunt | *Python, CTF challenge design, steganography, binary analysis, Git*

Sept. – Oct. 2025

- Designed binary reverse-engineering, PE/PCAP analysis, steganography, SSTI, and AES-ECB challenges for CruxHunt, an original Harry Potter-themed CTF at HackOWASP 8.0 (200+ participants), verified against a custom flag-validation pipeline

ACTIVITIES

OWASP Student Chapter TIET

Patiala, Punjab

Executive Member

Nov. 2025 – Present

- Contributed across Cyber, Marketing, and Design departments for HackOWASP 8.0, a national-level 24-hour hackathon with 200+ participants
- Competed in OWASP's XPLO(IT)2 CTF, solved 7+ PicoCTF challenges spanning web exploitation, cryptography, forensics, and reverse engineering, and exploited OWASP Juice Shop locally for hands-on vulnerability practice

Linux User Group, TIET

Patiala, Punjab

Core Member

Oct. 2025 – Present

- Participated in Linux and open-source workshops, collaborating on tooling initiatives to build CLI proficiency and system administration fundamentals